# Angular manual configuration changes from Angular 2 to Angular 19

## 🧭 Evolution of Angular Manual Configuration (2 → 19)

| Era | Main Config Files | Key Changes / Purpose |
|---|---|---|
| **Angular 2 (2016)** | • systemjs.config.js • tsconfig.json • main.ts bootstrap | - No CLI yet; setup done manually. - Developers configured SystemJS loader manually. - tsconfig.json had custom module: system or commonjs. - No angular.json, no build system abstraction. |
| **Angular 4 (2017)** | • angular-cli.json (early CLI) • tsconfig.json • environments/*.ts | - Angular CLI introduced! - CLI generated angular-cli.json for build & environment configs. - Developers no longer had to manage Webpack/SystemJS manually. |
| **Angular 5 – 7 (2017–2018)** | • angular.json *(replaced angular-cli.json)* • tsconfig.app.json, tsconfig.spec.json • polyfills.ts • .browserslistrc | - **angular.json schema introduced** for all project settings (build, serve, test, lint). - Split tsconfig.* files for app/test. - Introduced **AOT** and build configurations (production, development). - Polyfills separated into src/polyfills.ts. |
| **Angular 8 (2019)** | • angular.json (Webpack 4) • tsconfig.base.json • tsconfig.app.json | - Introduced **differential loading** → uses browserslist to generate ES5 + ES2015 bundles. - Ivy compiler preview toggle added in tsconfig.json: "angularCompilerOptions": { "enableIvy": true }. |
| **Angular 9 – 11 (2020–2021)** | • angular.json (Webpack 5) • tsconfig.app.json • polyfills.ts | - **Ivy became default**, enableIvy removed. - Added "aot": true default for all builds. - Added ngcc compatibility pipeline (for View Engine libs). - Optional "scripts"/"styles" arrays for global assets. |
| **Angular 12 – 14 (2021–2022)** | • angular.json • tsconfig.json with "strictTemplates": true | - Angular moved to **Webpack 5**. - Introduced "strict" and "strictTemplates" flags. - Deprecated es5BrowserSupport. - Added **inline style/image optimization** options. |

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| **Angular 15 (2022)** | • angular.json • Standalone component bootstraps in main.ts | - Introduced **Standalone APIs** (no NgModule). - Added "standalone": true usage. - Manual config changes: remove AppModule, use bootstrapApplication(). - CLI builds still Webpack 5. |
| **Angular 16 (2023)** | • angular.json (hybrid Webpack + esbuild preview) • tsconfig.app.json • Optional vite.config.ts (experimental) | - Optional builder: "@angular/build:browser" (esbuild). - New config keys for **signals** dev preview. - Still compatible with old Webpack builders. |
| **Angular 17 (2023 Nov)** | • angular.json simplified (new schema) • Uses "@angular/build" | - **Vite + esbuild default**. - Simplified build targets: build, serve, test. - Removed polyfills.ts (auto-injected polyfills). - Removed .browserslistrc (CLI uses built-in modern targets). |
| **Angular 18 (2024)** | • angular.json with unified targets (application, dev-server, ssr) | - **SSR integration** directly via "outputMode": "server". - "renderMode" added (Client / Server / Hybrid). - Removed older builders (browser, server, prerender). - Default TypeScript 5 config. |
| **Angular 19 (2025)** | • angular.json (minimal, unified) • tsconfig.json (simplified) | - **Full Vite + esbuild architecture.** - Manual config minimal: only "main", "index", "tsConfig", "renderMode". - Automatic polyfill injection, environment loading. - SSR and CSR merged; no need for separate server builds. |

## 🧩 File-by-File Evolution

| File | Introduced | Status (Angular 19) | Notes |
|---|---|---|---|
| **systemjs.config.js** | v2 | ❌ Removed | Replaced by CLI build config. |
| **webpack.config.js** | v2 (manual) | ❌ Removed | Managed by Angular CLI. |
| **angular-cli.json** | v4 | ❌ Removed | Replaced by angular.json (v6). |
| **angular.json** | v6 | ✅ Active | Core build configuration file. Simplified since v17. |

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|---|---|---|---|
| **tsconfig.json** | v2 | ✅ Active | TypeScript configuration. Modern Angular merges all nested tsconfig.*.json. |
| **polyfills.ts** | v5 | ⚠️ Optional | Mostly removed; polyfills handled automatically. |
| **.browserslistrc** | v6 | ⚠️ Optional | Superseded by internal browser targets in v17+. |
| **vite.config.ts** | v17+ | ✅ Optional | Used for advanced Vite customization (rare). |
| **server.ts / main.server.ts** | v8 | ✅ Active | Required for SSR / Hybrid rendering. |
| **app.routes.server.ts** | v18+ | ✅ New | Defines hybrid route modes for SSR + CSR. |

## 🧠 Key Manual Config Shifts

| Theme | Before (Angular 2–8) | After (Angular 17–19) |
|---|---|---|
| **Build setup** | Manual SystemJS/Webpack | Built-in via Vite + esbuild |
| **Compiler config** | AOT optional | Always AOT |
| **SSR config** | Manual Node/Express setup | Automatic via "ssr" builder |
| **Environment files** | environments/*.ts manual switching | Still supported but can use modern dynamic imports |
| **Polyfills** | Must list manually | Auto-detected per browser targets |
| **Browserslist** | Required for differential loading | No longer needed |
| **Testing** | Karma / Jasmine | Can opt for Vite test runner or Jest |
| **Hot reload** | Limited | Instant HMR via Vite |
| **Config complexity** | Dozens of keys | Only a few high-level entries |

## ⚙️ Example: Angular 19 Minimal Configuration

```
{
 "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
 "projects": {
  "my-app": {
   "projectType": "application",
   "sourceRoot": "src",
   "architect": {
    "build": {
     "builder": "@angular/build:application",
     "options": {
      "main": "src/main.ts",
      "index": "src/index.html",
      "tsConfig": "tsconfig.app.json",
      "outputPath": "dist/my-app"
     }
    },
    "serve": {
     "builder": "@angular/build:dev-server",
     "options": {
      "buildTarget": "my-app:build"
     }
    },
    "ssr": {
     "builder": "@angular/build:ssr",
     "options": {
      "outputMode": "server",
      "server": "src/main.server.ts",
      "routesFile": "src/app/app.routes.server.ts",
      "renderMode": "hybrid"
     }
    }
   }
  }
 }
}
```

| Version Range | Developer Manual Config Complexity | Tooling |
|---|---|---|
| **2 – 4** | ⚙️ High — manual SystemJS/Webpack setup | No CLI |
| **5 – 8** | 🔧 Medium — CLI Webpack, verbose JSON | @angular-devkit |
| **9 – 16** | 🧰 Stable — Ivy + Webpack 5 | Angular CLI |
| **17 – 19** | ✨ Minimal — Vite + esbuild unified | @angular/build |